7.2 Stretching Without Turning

WORKSHEET

The question

STEP III 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does it mean for T to leave a vector "unchanged in direction but multiplied by a scalar"? Write it as an equation.

STAGE 2 The image of the circle

Ask yourself To find the equation of E, express x and y in terms of X and Y (invert the matrix), and substitute into $x^2 + y^2 = 1$.

Ask yourself What shape is E, and what is its area? Use the eigenvectors.

STAGE 3 The greatest value of X

Ask yourself On C, $X = \frac{2}{5}(9x - 2y)$. How large can $9x - 2y$ be when $x^2 + y^2 = 1$?

STAGE 4 A tangent

Pause and try The point $\frac{1}{5}(3, 4)$ on C has tangent line $3x + 4y = 5$. A linear map sends tangent lines to tangent lines. Use this to find the tangent to E.

What's really going on

The spectral theorem

Every real symmetric matrix has a full set of perpendicular eigenvectors, with real eigenvalues. This is the *spectral theorem*, one of the central results of a first linear algebra course. It says that every symmetric linear map is, in suitable perpendicular coordinates, just a stretch along each axis. Equivalently, every quadratic form like $8X^2 + 12XY + 17Y^2$ can be rotated into the form $\lambda_1 u^2 + \lambda_2 v^2$, which is why every central conic is an ellipse or hyperbola with perpendicular axes (the *principal axis theorem*). The idea goes back to Cauchy in the 1820s, who used it to find the principal axes of quadric surfaces and of the stress in a solid body.

Symmetry is what makes this work. A matrix that is not symmetric need not be so obliging: a rotation through 90° turns every direction, so it has no real eigenvectors at all, and the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only one eigenvector direction, so it is not a stretch along any pair of axes.

Eigenvectors everywhere

Here are a few of the places eigenvectors appear.

- **Vibrations.** A system of masses and springs vibrates in *normal modes*, patterns in which every part moves in step at a single frequency. These are eigenvectors of a matrix, and the frequencies come from the eigenvalues (Section 11.2).
- **Data.** The directions in which a cloud of data points is most spread out are the eigenvectors of its covariance matrix. Finding them is *principal component analysis*, one of the basic tools of statistics and machine learning (Section 13.7).
- **Quantum mechanics.** The possible results of measuring a quantity are the eigenvalues of a matrix (or operator), and the energy levels of an atom are eigenvalues.
- **Web search.** Google's PageRank, the original idea behind the search engine, ranks pages by the entries of the eigenvector, with eigenvalue 1, of a huge matrix describing the links between pages.

Try this next

1. **Finding eigenvalues.** Show that the eigenvalues of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are the roots of $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$, and check that the matrix in the question gives $\lambda = 2$ and 4.

2. **Perpendicular eigenvectors.** Show that if $\mathbf{M}$ is symmetric and $\mathbf{Mu} = \lambda\mathbf{u}$, $\mathbf{Mv} = \mu\mathbf{v}$ with $\lambda \neq \mu$, then $\mathbf{u} \cdot \mathbf{v} = 0$. (*Hint:* compute $\mathbf{v} \cdot \mathbf{Mu}$ in two ways.)

3. **A rotation.** Show that the rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ has no real eigenvectors unless θ is a multiple of π. What are its complex eigenvalues?

4. **Powers.** Using the eigenvectors, find $T^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. (*Hint:* write $(1, 0)$ as a combination of $(1, 2)$ and $(2, -1)$.)

Further reading

- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 6 is an excellent introduction to eigenvalues and their applications.
- **Kurt Bryan & Tanya Leise, "The $25,000,000,000 eigenvector: the linear algebra behind Google"**, *SIAM Review* 48 (2006). Readable and freely available.
- **3Blue1Brown, *Essence of Linear Algebra*** (video series, free online). Beautiful animations of exactly the geometric ideas in this section.

The full guided solution is in Section 7.2:
https://beyondstep.keithmcnulty.org/ch07/eigenvectors.html